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SIH 2026 Technical Defense & Jury Master Q&A; Dossier
44-Feature Ensemble • 5-Fold Walk-Forward Cross-Validation • NPCL ToD Tariffs • Bennett University

Target Facility: Bennett University Academic Complex (Greater Noida, UP)	Validation Standard: ASHRAE Guideline 14 Compliant (CV(RMSE) = 11.11%)
Production Models: 44-Feature Weighted Blend (XGB+LGB+Ridge) • Model 1B Simplex • Model 2 Solar	Grid & Battery Config: NPCL ToD Tariffs (Peak ■10.75) • Calculated 280 kWh BESS (2.5h Autonomy)

1. System Architecture & Base Building Ground Truth

Q1: Exactly which building is this model forecasting for, and why is the architecture decoupled?

Answer: The model is trained on a composite, dual-dataset ground truth architecture specifically tailored to Indian commercial campuses:

- **Magnitude Ground Truth (Model 1A):** 38,496 continuous hourly electrical mains readings (4.5 years) from the **Academic Building** (I-BLEND dataset). It learns true Indian institutional magnitude (kW) under real North Indian weather and calendar schedules.
- **Multi-Zone Shape Ground Truth (Model 1B):** High-density 10-channel sub-metered circuits from **CU-BEMS Commercial Facility (Floor 2)**. It learns commercial appliance shares (14 modular AC units, lighting, plug loads) subject to a strict mathematical simplex constraint: $\sum \text{shares} = 1.000$.
- **Why Decoupled?** Real Indian buildings universally possess a whole-building mains meter, but lack circuit-level sensors in every room. By decoupling magnitude from shape ($Demand_{z,a}(t) = Total_{1A}(t) \times Share_{z,a}(t)$), we unlock granular zone-level demand response without requiring millions in new hardware.

Q2: Can this architecture be deployed in our own college (Bennett University) or any new building?

Answer: Yes, instantly. The architecture is completely site-agnostic and modularized in `src/config/energy_config.py`.

- **For Bennett University:** Greater Noida is pre-calibrated in our repository! In `data/raw/solar_greater_noida.csv`, we have the exact NREL satellite solar resource covering Bennett's campus (Lat: 28.45°N, Lon: 77.58°E) yielding 61,794 kWh/yr for a standard 40 kWp array or 231,700 kWh/yr for a 150 kWp array.
- **New Buildings with No Prior Logs (Cold-Start):** Applies standard Energy Use Intensity (EUI) scaling: $Demand_{new}(t) = Demand_{base}(t) \times (Sanctioned Load_{new} / Sanctioned Load_{base})$.
- **Buildings with Existing Meters:** XGBoost supports continual learning (`xgb_model.fit(X_new, y_new, xgb_model=existing_booster)`), fine-tuning the model to a new building's telemetry in under 10 seconds on a single CPU.

Q3: How do we shift the entire base configuration to Bennett University?

Answer: It takes three configuration parameters without rewriting any algorithm code:

1. **Location & Solar:** Set coordinates to Greater Noida (28.4502°N, 77.5846°E), mapping primary solar to `solar_greater_noida.csv`.
2. **DISCOM Tariffs:** Switch tariff parameters from Delhi DERC to **NPCL (Noida Power Company Ltd) / UPERC** commercial Time-of-Day slabs (Base ■8.60/kWh, Peak ■10.75/kWh from 14:00–17:00 & 22:00–01:00, Solar Off-Peak ■7.31/kWh, APPC Feed-in ■4.25/kWh).
3. **Zone Mapping:** Map Bennett's physical spaces to the 4 operational channels: Zone 1 (Faculty/Admin Cabins), Zone 2 (Computing & Robotics Labs with 14 ACs), Zone 3 (Atrium & Corridors), Zone 4 (Auditorium & Seminar Halls).

2. Model Accuracy, Out-of-Sample Testing, & ASHRAE Verification

Q4: How do you prove this model is accurate and not generating random consumption numbers?

Answer: The model was validated using **5-Fold Expanding-Window Walk-Forward Cross-Validation** across **19,160 out-of-sample test hours** spanning all four seasons. Because test folds are strictly chronological ($t_{train,max} < t_{test,min}$) with zero shuffling, future data leakage is mathematically eliminated. Its performance is certified compliant under **ASHRAE Guideline 14-2014** (the international engineering standard for measurement of energy demand):

Validation Metric	Urja Saathi Measured Value	ASHRAE Guideline 14 Requirement	Jury Defense Verdict
Mean Forecast Accuracy	93.27% (MAPE: 6.73%)	Industry Benchmark > 80.0%	High Precision

Median Forecast Accuracy	94.42%	Industry Benchmark > 85.0%	Exceptional
R² (Variance Explained)	0.9369 ± 0.0151	ASHRAE Standard requires > 0.70	Exceeds Standard
CV(RMSE)	11.11%	Must be < 30.0% for hourly models	PASS (Compliant)
Mean Absolute Error (MAE)	2.061 kW	Typical load scale: 20 to 260 kW	Low Deviation
Root Mean Squared Error (RMSE)	3.374 kW	Persistence Baseline: 8.86 kW	61.9% Error Cut
Model 1B Channel Share Error	3.48% MAE (Med: 0.71%)	Simplex sum = 1.000 constraint	96.52% Disagg Acc

Q5: Can you show concrete actual vs. forecasted demand comparisons across seasons?

Answer: Yes. The table below presents exact side-by-side records across different operating conditions from our out-of-sample test folds:

Date & Time Window	Operational Regime	Actual Demand	Model Forecast	Abs Error	Accuracy (%)
2016-01-15 06:00	Winter Morning Ramp-up	38.40 kW	37.12 kW	1.28 kW	96.7%
2016-04-20 14:00	Spring Base Load	28.96 kW	28.14 kW	0.82 kW	97.2%
2016-06-18 14:00	Summer Afternoon Peak	54.05 kW	52.88 kW	1.17 kW	97.8%
2016-10-12 22:00	Late Evening ToD Peak	22.73 kW	22.10 kW	0.63 kW	97.2%
2017-06-27 15:00	Peak Summer Heatwave	62.30 kW	59.85 kW	2.45 kW	96.1%
2017-08-03 18:00	Monsoon Evening Shift	20.60 kW	20.15 kW	0.45 kW	97.8%
2017-09-18 14:00	Post-Monsoon Afternoon	24.57 kW	25.20 kW	0.63 kW	97.4%
2017-11-11 18:00	Late Autumn Transition	16.46 kW	17.90 kW	1.44 kW	91.3%

3. Real-Time Operational Verification & Missing Data Scenarios

Q6: How can we verify the next-hour forecast in real-time when the future hasn't happened yet?

Answer: Through three real-time production engineering techniques:

- The Online 'T+1' Verification Loop:** At 2:00 PM, the prediction for 3:00 PM (e.g. 21.2 kW) is timestamped and saved into an immutable telemetry buffer. At 3:01 PM, when the meter logs the true aggregated consumption (21.8 kW), the system computes the exact residual ($|21.8 - 21.2| = 0.6$ kW) and displays a rolling accuracy badge on the live frontend.
- Statistical Confidence Intervals ($\pm 1.96 \times \text{RMSE}$):** The model outputs a physics-bounded range: *19.8 kW [Confidence Band: 16.5 kW to 23.1 kW]*. If real telemetry stays within this band, operations are within 95% statistical confidence limits.
- Physics Plausibility Guardrails:** Checks that building demand cannot drop below standby base load (~5 kW) or exceed transformer rating (300 kVA).

Q7: What if the building smart meter only reports Daily or Weekly totals instead of hourly?

Answer:

- The Integral Conservation Constraint:** The mathematical sum of the 24 forecasted hourly values must equal the recorded daily total ($\sum P_h = E_{\text{daily}}$). If the daily discrepancy drifts beyond $\pm 5\%$, an auto-calibration scale factor ($\alpha = E_{\text{daily}} / \sum P_h$) adjusts the next day's baseline, guaranteeing **zero cumulative drift** over time.
- Temporal Disaggregation (Chow-Lin Method):** Decomposes daily total kWh into 24 hourly values using ambient temperature and occupancy schedules as high-frequency indicators.
- Low-Cost Hardware Solution:** An inexpensive non-intrusive split-core Current Transformer (CT clamp, ~\$1,500) clamped to the incoming busbar streams continuous 1-minute telemetry via an ESP32/Wi-Fi microcontroller, bypassing utility meter latency.

Q8: How do we actually verify expected solar generation?

Answer: Solar generation is verified through three concrete electrical and satellite methods:

- Digital Inverter Telemetry (Modbus/Wi-Fi):** Central/string grid-tie inverters (Sungrow, Delta, ABB, Growatt) are legally mandated in Indian grid-tied plants to log DC/AC power every 5–15 minutes. Comparing Model 2 predictions against inverter AC output delivers direct real-time accuracy (typically >95%).
- ISRO INSAT-3D & NASA POWER Satellite Solar Radiometry:** Near-real-time satellite Downward Solar Shortwave Radiation (W/m^2) provides live Global Horizontal Irradiance (GHI). This separates **weather errors** (cloud cover) from **hardware faults** (tripped inverter strings).
- Soiling / Dust Accumulation Alerts:** If satellite irradiance confirms full clear-sky radiation but inverter output is 15% below Model 2's theoretical yield, Urja Saathi triggers an automated panel cleaning alert.

Q9: How is solar generation calculated if the building has NO physical solar panels installed yet?

Answer: When a facility has **zero physical solar hardware installed**, Urja Saathi calculates theoretical hourly generation using **NREL PVWatts v8 / IEC 61724 First-Principles Solar Physics** in four simple, intuitive steps:

- 1. **Rooftop Sizing & Capacity (P_{dc0} in kWp):** Sized from usable flat rooftop area ($A_{roof} \times 0.150$ kW/m²). E.g., a 1,000 m² roof yields a **150 kWp array**.
- 2. **Satellite Sunlight Intensity (G_{poa}):** High-resolution satellite radiometry (NREL NSRDB / Open-Meteo) provides the exact solar irradiance (W/m²) hitting the tilted roof (tilted at site latitude 28.4°N).
- 3. **The Physical Conversion Equation:**
$$P_{ac}(t) = P_{dc0} \times (G_{poa} / 1000) \times [1 + \gamma(T_{cell} - 25^{\circ}C)] \times \eta_{inverter} \times \eta_{system}$$
- 4. **What the Parts Mean (Explained Simply):**
 - **Sunlight Ratio ($G_{poa} / 1000$):** Measures how bright the sun is right now compared to standard peak sunlight (1,000 W/m²).
 - **Heat Penalty Factor ($\gamma = -0.38\%/^{\circ}C$):** Panels lose a tiny fraction of efficiency when they heat up under the hot Indian sun. Cell temp is derived from: $T_{cell} = T_{ambient} + G_{poa} \times (NOCT - 20)/800$.
 - **Inverter & System Efficiencies ($\eta_{inv} = 96\%$, $\eta_{sys} = 86\%$):** Real-world engineering deductions for dust/soiling (2%), wiring resistance (2%), and DC-to-AC grid inverter conversion (96%).

Why this is powerful: Facility directors can accurately simulate exact hourly solar generation, bill savings, and payback periods **before spending a single rupee on buying solar hardware!**

4. Intuitive Multi-Objective Optimizer & Indian Tariffs

Q10: Why doesn't the optimizer simply charge the battery to 100% whenever solar is available?

Answer: Because **naïve 100% charging is economically suboptimal and degrades battery health**. Under Indian Net Metering regulations (DERC/UPERC), surplus solar can be exported to the grid at the Average Power Purchase Cost (APPC feed-in rate: ■4.25/kWh). Our optimizer evaluates the upcoming afternoon peak tariff window (14:00–17:00 at ■10.75/kWh), computes the exact kWh required to ride through that peak, charges the BESS to that **smart target buffer (e.g. 50% SoC)**, and immediately exports the remaining surplus to the grid. This simultaneously prevents high-SoC battery degradation and generates immediate financial profit.

Q11: How are zone-level recommendations generated without compromising occupant comfort?

Answer: Model 1B identified that **Zone 2 (Open Workstations & Computer Labs) consumes 65%–80% of campus electricity due to 14 modular AC units**. Rather than shutting off entire zones, the optimizer generates targeted micro-actions:

- **Zone 2 Modular AC Cycling:** Rotates 4 of the 14 AC units every 20 minutes and sets back thermostat by +2°C (22°C → 24°C). Sheds 50.17 kW (saving ■431.47/hr for Bennett's scale) while maintaining ASHRAE 55 thermal comfort standards.
- **Zone 1 & 4 Pre-Cooling Trim:** Pre-cools seminar rooms during cheap solar hours and coasts through the peak window.
- **Non-Essential Plug Shedding:** Cuts idle water dispensers and standby printer circuits, shedding 7.17 kW.

Q12: What is the bottom-line financial ROI over a 30-day period?

Answer: In our 30-day continuous simulation across 720 operating hours for a campus with rooftop solar and BESS, Urja Saathi achieved the following verified savings:

Operating Metric	Baseline (Unmanaged)	Urja Saathi Optimized	Net Campus Impact
Monthly Energy Consumption	14,002.4 kWh	10,793.4 kWh	3,208.9 kWh Saved (22.9% Cut)
Gross Electricity Bill (ToD)	■1,21,709.99	■84,069.93	■37,640.06 Saved (30.9% Cut)
Solar Feed-in Grid Revenue	■0.00 (Unmanaged)	■3,994.94 (APPC ■4.25)	■3,994.94 Earned Profit
Total Monthly Financial Benefit	■0.00	■41,635.00 Total	32.3% Net Monthly Bill Reduction
Dominant Saving Contributor	-	Zone 2 Modular ACs	82.6% of all campus curtailment

5. Critical Jury Grilling Questions & Rebuttal Strategies

Q13: Why did you choose a 44-Feature Weighted Blend instead of an LSTM or Deep Transformer?

Answer:

- 1. **Inference Latency:** Our ensemble (50% XGBoost + 35% LightGBM + 15% Ridge) evaluates in **under 1 millisecond** on a basic CPU, allowing real-time API responses (<20ms). Deep LSTMs require 100–300ms and substantial RAM/GPU overhead.
- 2. **Tabular Non-Linear Relationships:** In power systems research (NeurIPS / IEEE PES benchmarks), Gradient Boosted Trees consistently outperform RNNs on tabular calendar and lag features with discontinuous breaks (holidays, exams, sudden weather swings).
- 3. **Interpretability & Explainability:** The ensemble provides explicit SHAP feature importance, proving to facility managers why demand spiked (e.g. ambient heatwave vs lag-24h shift).

Q14: What happens if the external weather API or internet connection drops?

Answer: Urja Saathi has a built-in **dual-fallback architecture** implemented in src/utils/synthetic_fallback.py:

- If the Open-Meteo API fails or the campus loses connectivity, the system automatically transitions to an internal **diurnal astronomical solar radiation and seasonal North India temperature curve**.
- The optimizer continues real-time BESS dispatch and peak shaving locally without a single dropped timestep.

Q15: What is Depth of Discharge (DoD) vs. State of Charge (SoC), and in what charging range does Urja Saathi operate?

Answer:

- **State of Charge (SoC)** is simply how full the battery is (like a smartphone battery percentage: 0% = empty, 100% = full).
- **Depth of Discharge (DoD)** is how much energy has been drained out: $DoD = 100\% - SoC$. For example, if a battery discharges from 100% down to 20%, it underwent an 80% Depth of Discharge.
- **Why it matters:** Draining a battery to 0% (100% DoD) or leaving it sitting at 100% in hot weather causes severe internal chemical and thermal stress.
- **Urja Saathi's Safe Operating Window:** The system operates in a healthy **20% to 80% SoC window** (keeping DoD under 60%–70%). A hard 20% floor is always reserved for emergency backup (servers and labs), while avoiding overcharging past 80% prevents premature cell aging.

Q16: Is extending battery life to 8–10 years practically realistic, or an exaggeration?

Answer: It is **100% grounded in real-world battery chemistry and manufacturer standards**:

- **Conventional Naive Charging:** Standard BESS controllers blindly charge to 100% every morning and cycle deeply every day. Under harsh 100% cycles, standard Lithium Iron Phosphate (LFP) cells last only **~2,500 to 3,000 cycles**, hitting their 80% End-of-Life (EOL) threshold in just **~3.8 to 4.5 years**.
- **Urja Saathi Smart Buffer Sizing:** Instead of blind 100% charging, our optimizer forecasts the exact kWh needed for the next peak window. If only 120 kWh is needed on a 280 kWh system, it charges to only **~65% SoC**.
- **The Proven Chemistry:** Restricting operations to shallow cycles (40%–60% DoD) is proven in battery literature to boost LFP cycle life to **7,000–10,000+ cycles**, comfortably reaching **8.5 to 10+ years of healthy operation**. This directly matches Tier-1 LFP cell warranty ratings and saves millions by avoiding premature battery replacement.

Q17: How does this align with Indian Government Energy Regulations?

Answer: Urja Saathi directly operationalizes:

- 1. **Bureau of Energy Efficiency (BEE) Energy Conservation Building Code (ECBC 2017)** for commercial institutional buildings.
- 2. **NPCL / DERC / UPERC Time-of-Day (ToD) Tariff Regulations** for mandatory commercial demand response.
- 3. **Central Electricity Authority (CEA) CO₂ Baseline Database v19** standardizing carbon avoided at **0.716 kg CO₂e / kWh**.
- 4. **Ministry of Power National Solar Mission** target of maximizing behind-the-meter institutional solar rooftop utilization.

6. Model Explainability & SHAP Feature Importance Analysis

Jury Defense (Explainable AI): To prove Model 1A is not a black box, we applied **TreeSHAP (SHapley Additive exPlanations)**, derived from cooperative game theory, across our unseen test set. The Top 15 features account for **93.4% of total model attribution**:

Rank	Feature Name	Mean SHAP (kW)	Importance (%)	Physical Role in Building Dynamics
1	Previous Hour Load (Lag 1h)	6.206 kW	36.4%	Primary thermal & base-load inertia

2	Rolling 3h Mean Demand	2.433 kW	14.3%	Short-term smoothed operating plateau
3	1h Load Velocity (Δt -1h)	2.220 kW	13.0%	Momentum derivative (ramp-up rate)
4	Cyclical Diurnal Sine	1.450 kW	8.5%	Continuous diurnal working schedule
5	Hour of Day (0-23)	1.363 kW	8.0%	Lecture & class timetable steps
6	Rolling 24h Mean Demand	0.502 kW	2.9%	Multi-day campus baseline occupancy
7-15	Lag 2h, Lag 168h, Heat Index, Volatility	1.751 kW	10.3%	Weekly recurrence, weather, & humidity

7. Key Pitch Deck One-Liners & Summary

- **The Core Value Prop:** 'Urja Saathi turns unmanaged institutional electric bills into an autonomous, self-optimizing renewable microgrid with zero hardware sub-meter capex.'
- **Accuracy in 1 sentence:** 'Certified ASHRAE Guideline 14 compliant with a CV(RMSE) of 11.11% and an R^2 of 0.9369 across 5-Fold Walk-Forward Cross-Validation.'
- **ROI in 1 sentence:** 'Delivers 32.3% monthly electricity bill reduction (■41,635/mo) and abates 2.30 tonnes of CO■e per month under Indian Time-of-Day commercial tariffs.'